

A scenic view of a Stockholm street scene featuring historic European architecture, a canal with swans, and a blue gradient overlay on the right side.

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Probability of Causation for Lung Cancer Compensation: A Scoping Review of Methods and Alternatives

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Introduction

- A Dutch government-funded no-fault compensation scheme for serious substance-related occupational diseases (TSB), was established in 2023.
- One-time financial compensation (€27.030) for affected workers.
 - Lung cancer due to occupational asbestos and silica
- *Extension to other lung carcinogens (IARC Monographs 1-141):*
 - 20 agents
 - 8 occupations, industries or processes

Probability of Causation (PoC)

General causation

"Can asbestos cause lung cancer?"

What would happen if a healthy person is exposed?

Epidemiological studies at the population level

Specific causation

"Did asbestos cause this worker's lung cancer?"

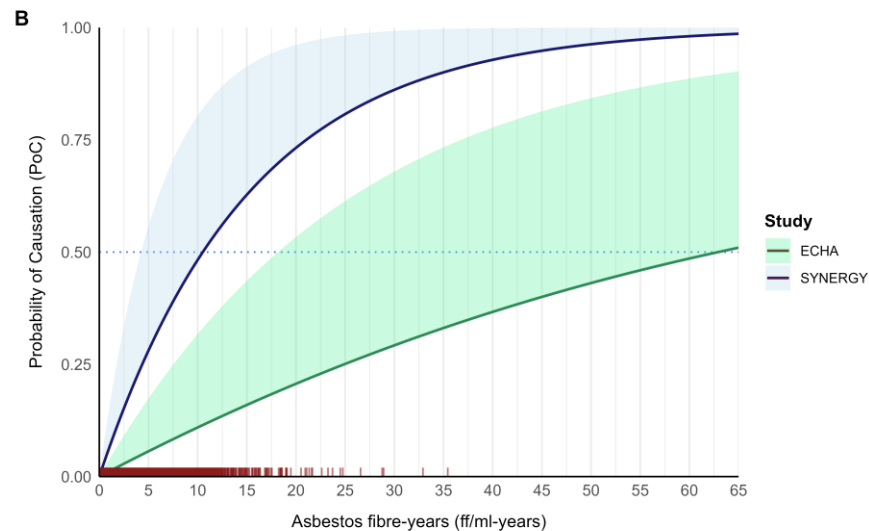
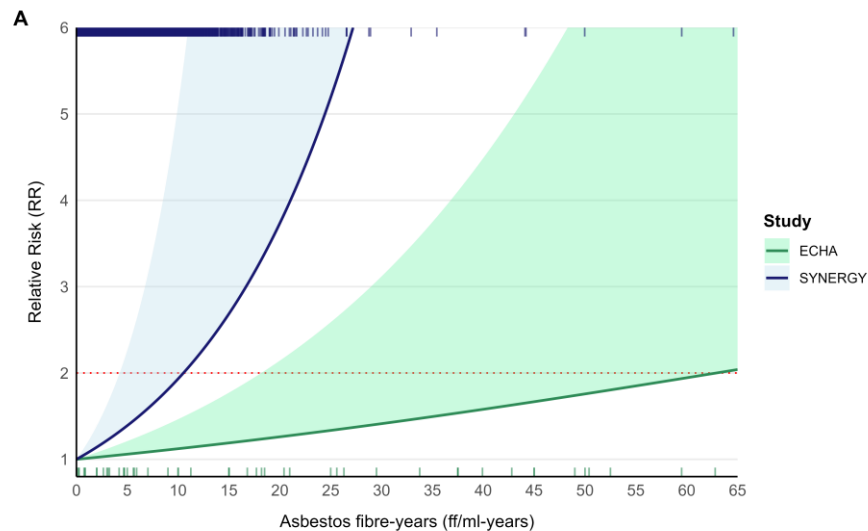
Would the cancer not have occurred without the exposure?

Counterfactual quantity at the individual level

If $\text{PoC} \geq 50\%$, the exposure is "more likely than not" the cause → compensation threshold

$$\text{PoC} = \frac{\text{Risk}_{\text{exposed}} - \text{Risk}_{\text{unexposed}}}{\text{Risk}_{\text{exposed}}} = \frac{\text{RR} - 1}{\text{RR}}$$

PoC: Lung cancer due to asbestos



Limitations of the PoC

- 1** Strong causal assumptions
- 2** Population averages $\neq$ individual risk
- 3** Exposures assumed uniform
Studies often treat workers as homogeneously exposed groups.
- 4** Ignores acceleration of disease
PoC does not account for earlier disease onset (accelerated by occupational exposure)
- 5** Multiple sources of uncertainty
Statistical error, exposure measurement error, and applying population data to individuals all convey uncertainty.

Objectives

- Describe the literature on probability of causation (PoC) estimation for lung cancer
→ **PoC Review**
- Scope for alternative methods to the PoC for multi-causal disease compensation, discussing advantages and limitations
→ **Alternatives Review**

Methods: PoC Review

1  Scope	2  Database Search	3  Machine-assisted screening	4  Study Selection
<i>Pilot (Dec 2024)</i> - Population: workers at risk of lung cancer Concept: PoC from quantitative exposure-response relations Context: lung cancer	<i>Final Search (May 2025)</i> - PubMed, Embase and OpenAlex - Backward and forward citation retrieval (Aug 2025) 1,715 unique records	<i>ASReview</i> - Assisted-learning models - Multilingual embeddings for non-English records 184 records screened 59 retrieved for full-text	<i>Excluded</i> - No quantitative exposure-response (n = 14) - Not lung cancer (n = 12) - Not PoC (n = 7) 15 epidemiological studies 12 theoretical/review

Methods: Alternatives Review

1  Scope	2  Database Search	3  Screening	4  Study Selection
Population: workers with occupational disease Concept: Types of damages and reparation, including pecuniary and immaterial damages. Context: Workers' compensation systems	<i>Final Search (May 2025)</i> • PubMed, Embase and CINAHL • + studies identified in the PoC review 257 unique records + 18 from PoC review	<i>All records screened</i> 37 + 18 retrieved for full-text	<i>Excluded</i> • Full-text NA (n = 13) • Multi-causality not discussed (n = 14) • Not an alternative method for compensation (n = 12) 4 + 8 selected

Results

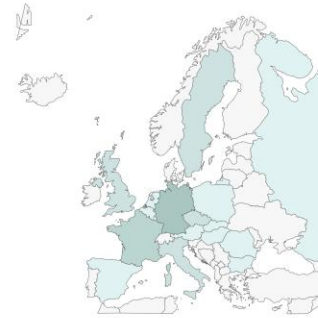
PoC review

- 15 epidemiological studies included (1984–2025)
 - Radiation most assessed (n = 11), then asbestos (n = 3), and coal tar pitch volatiles (n = 1)
 - 13 studies used PoC threshold of 0.5
 - Upper 95% (n=8) or 99% (n=2) CIs used to favour claimants
 - Main uncertainties: statistical error (n=10), population risk transfer (n=7), measurement error (n=6)
 - Uncertainty distributions propagated through Monte Carlo simulation in more recent studies
 - Lack of consensus between additive and multiplicative effects models. Some use model averaging techniques.
- Counterintuitive compensation decisions (Armstrong & Thériault, 1996)

Americas



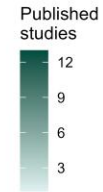
Europe



Asia



Oceania

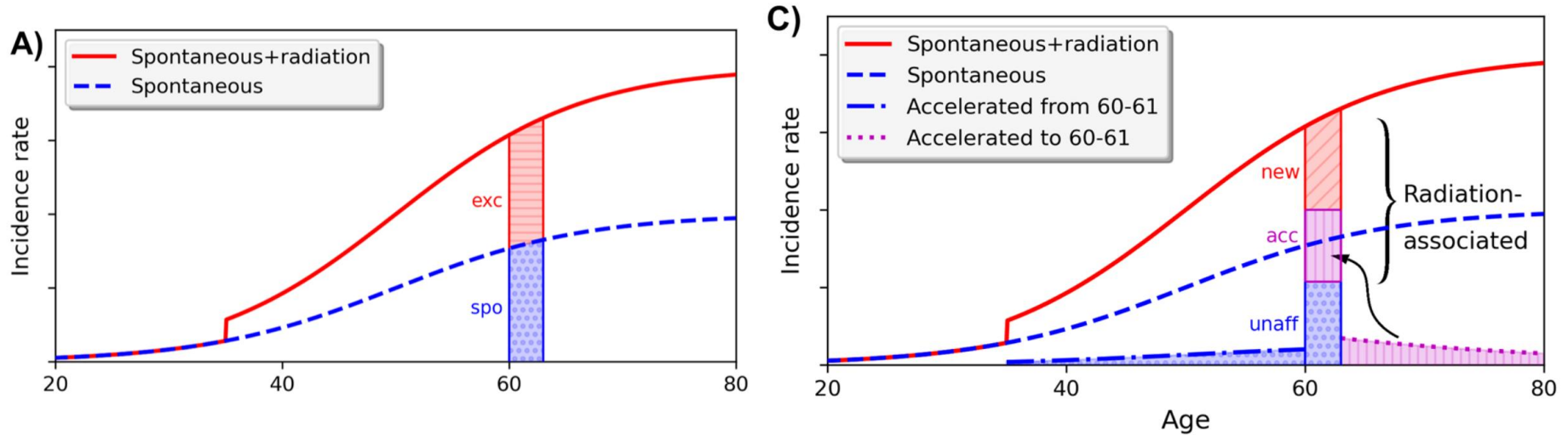


Map of countries of study participants (for research articles) and authors (for theoretical and review articles). No studies (or authors) based in Africa were found.

Alternatives review

- 12 studies included
- Mechanistic models and clonal expansion models (n = 4)
 - ✓ Acceleration; aligned with PoC
 - ✗ Difficult interpretation and sensitive to spec.
- Broader harm dimensions beyond binary eligibility (n = 6)
 - ✓ Better alignment with legal damage assessments
- Years of life lost attributable to exposure (n = 2)
 - ✓ Acceleration; compensation proportional to damage
 - ✗ Estimation requires specific cohort characteristics

Probability of Harm based on TSCE



Years of life lost due to exposure

1 Individual PoC

$$\text{PoC}(t) = [h_1(t) - h_0(t)] / h_1(t)$$

Probability that a death at time t was caused by the exposure. h_0 is counterfactual, so PoC is **not identifiable**.

2 Weighted by time lost

$$y(t) = \text{PoC}(t) \times E[T - t \mid T > t]$$

Zero if outcome not caused by exposure; otherwise the remaining life expectancy the person would have had without it. **Not identifiable**.

3 Aggregate over the population

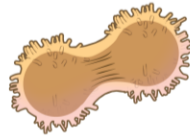
$$\text{EYLL} = E[T_0] - E[T_1]$$

The expectation of the sum of individual contributions is the contrast in mean survival without vs. with exposure.

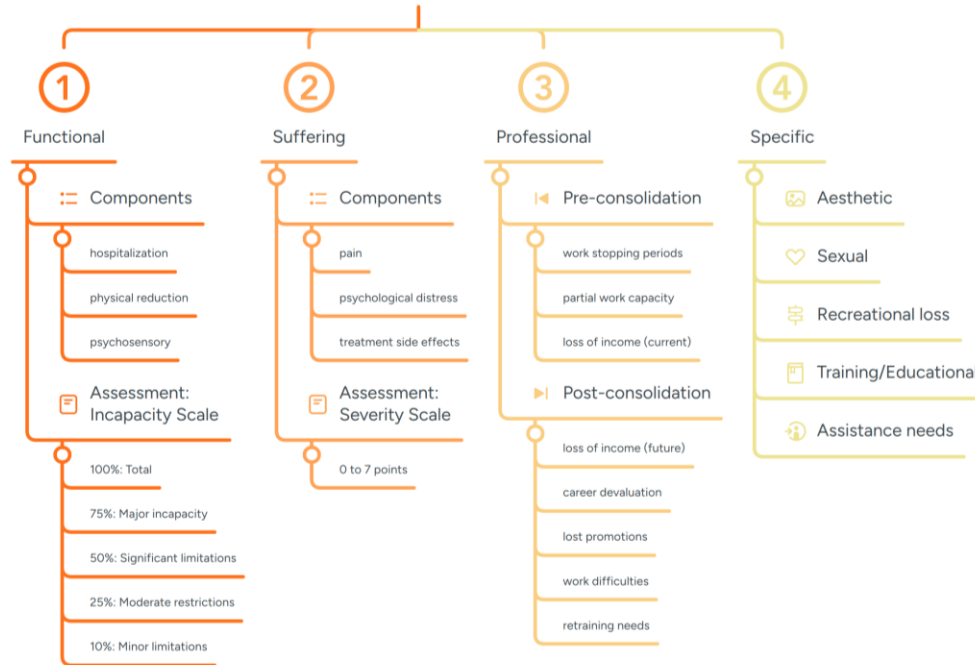
4 Average expected years of life lost

$$\text{EYLL} = \int (S_0(t) - S_1(t)) dt$$

The area between the two survival curves. Determined entirely by population-average survival is **identifiable** from cohort data.



Cancer Impacts and Damages



Discussion

- Strengths:
 - Protocol registration and open science
<https://github.com/UtrechtUniversity/PoC-scoping-review>
 - Systematic approach to describe PoC methods and alternatives
- Limitations:
 - PoC review restricted to lung cancer
 - Screening with assisted learning conveys risk of not identifying relevant studies. Risk minimized by using different models and screening rounds.
 - Automated translations obtained for a subset of studies: Japanese (n = 2), French (n = 1), Mandarin (n = 1), and Portuguese (n = 1)
 - Alternatives review was exploratory

Conclusions

- Uncertainty quantification has been central when applying PoC in compensation scheme contexts, to favor claimants.
- PoC improvements accounting for acceleration (e.g., probability of harm) interesting but preliminary.
- Alternative methods proposed many decades ago (years of life lost) have seen little uptake, likely due to data requirements.
- Compensation based on other measures of harm and not PoC may be justifiable and better aligned with harm interpretations in tort law.

Thank you!

Collaborators

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Questions?

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